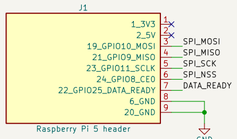
Karetao bench wiring: STM32F405 hub, TCA9548A mux, AS5600 encoders, CAN to six MKS drives, SPI to the Raspberry Pi 5.

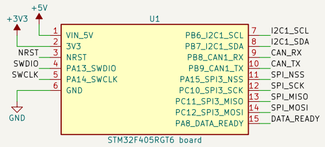
Signals are connected by net labels of the same name. Power symbols: +3V3, +5V, GND, VMOT_SW (motor supply after the E-stop).

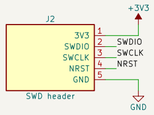
Confirmed by the documents: STM32 pins (ARM_SENSOR_HUB.md), mux address 0x70 and axis N on channel N+2, encoder fitted per axis, drive models and CAN IDs 0x001 to 0x006 (SYSTEM_CONFIGURATION.md 2.1, 2.2), Pi header pins 19/21/23/24/22 (Pi README), 12 to 24 V drive supply (MKS manual).

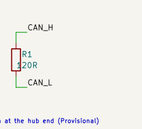
Provisional (not stated in any document): STM32 board type and its 5 V input; TJA1050 S pin tie; I2C pull-up values and location: mux RESET pull-up (no MCU line, FW-12); AS5600 DIR tied low; termination placement; motor supply voltage; the E-stop itself (SAFE-01, ruling R25b: to be fitted); drive switch and motor phase wiring (Arctos drawing, bench check). Decoupling on breakout boards is not drawn. Gripper parts (Nano, MCP2515, servo) are out of scope and not drawn (charter).

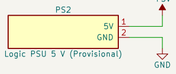
Generated by Hardware/bench-wiring/generate_schematic.py: edit the script, not this file, then regenerate. Run ERC in KiCad after any change.











R1: termination at the hub end (Provisional)

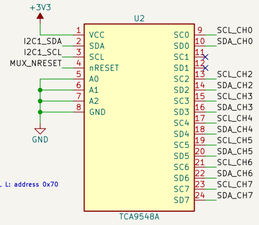
U10 TJA1050 on +5V: RXD swings to 5 V into PB8 (FT, 5 V tolerant only while the STM32 is powered; no pull-up). S tied low (Provisional).

CAN bus: CAN_H and CAN_L daisy-chained U10 -> M1 ... M6 (and J3 when used), 120R at each end (R1, R2); every drive's own terminator off (57D DIP 1, 42D shorting cap). Drives and adapter share GND with the hub (MKS manual p. 18). Drive blocks: bottom of the sheet.

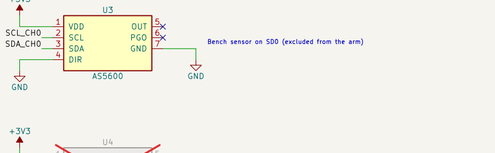
J3: USB-CAN adapter for bring-up phase 7a. Its GND is connected (the Arctos drawing leaves it floating).



R3, R4: I2C pull-ups; R5: mux reset pull-up. Values and location Provisional.

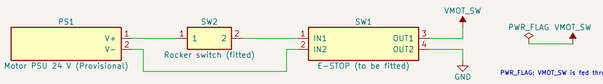


A2 A1 A0 = L L L: address 0x70





R2: far-end termination (Provisional)



Drive I/O, from the original Arctos wiring. Home and limit switch wiring and motor phase order are Provisional until confirmed on the bench (BRING_UP phase 1).

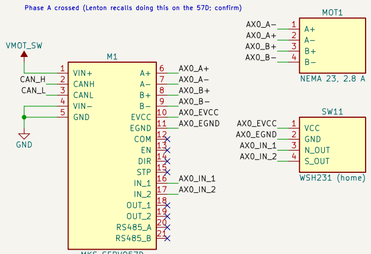
Phase A is crossed at every drive (drive A+ to motor A-), as in the Arctos drawing; it only sets the turning direction. Switch modules take 5 V from the drive's EVCC/EGND (20 mA budget).

Axis 0: Joint X (Base)

CAN ID 0x001, SERVO57D; motor NEMA 23, 2.8 A

Switches on IN_1 / IN_2: DIP 2, 3 ON: EVCC 5 V, DIP 1 (terminator) OFF.

Phase A crossed (Lenton recalls doing this on the 57D; confirm)

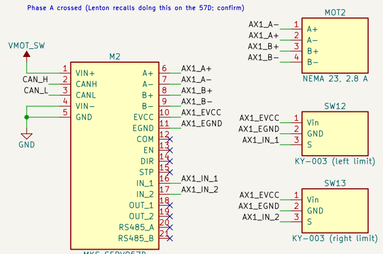


Axis 1: Joint Y (Shoulder)

CAN ID 0x002, SERVO57D; motor NEMA 23, 2.8 A

Switches on IN_1 / IN_2: DIP 2, 3 ON: EVCC 5 V, DIP 1 (terminator) OFF.

Phase A crossed (Lenton recalls doing this on the 57D; confirm)

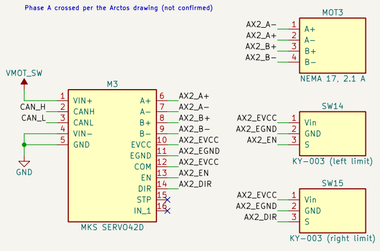


Axis 2: Joint Z (Elbow)

CAN ID 0x003, SERVO42D; motor NEMA 17, 2.1 A

9EH port mapping on: left -> En, right -> Dir; COM to EVCC. IN_1 unused. Terminator cap off.

Phase A crossed per the Arctos drawing (not confirmed)

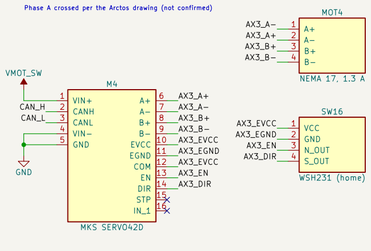


Axis 3: Wrist Joint 1 (Wrist 1)

CAN ID 0x004, SERVO42D; motor NEMA 17, 1.3 A

9EH port mapping on: left -> En, right -> Dir; COM to EVCC. IN_1 unused. Terminator cap off.

Phase A crossed per the Arctos drawing (not confirmed)

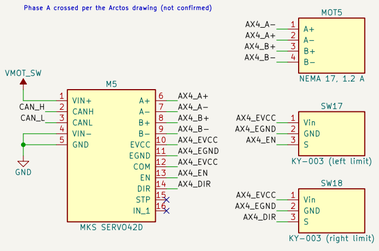


Axis 4: Wrist Joint 2 (Wrist 2)

CAN ID 0x005, SERVO42D; motor NEMA 17, 1.2 A

9EH port mapping on: left -> En, right -> Dir; COM to EVCC. IN_1 unused. Terminator cap off.

Phase A crossed per the Arctos drawing (not confirmed)

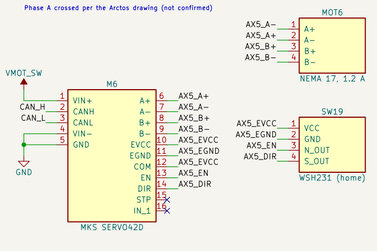


Axis 5: Wrist Joint 3 (Wrist 3)

CAN ID 0x006, SERVO42D; motor NEMA 17, 1.2 A

9EH port mapping on: left -> En, right -> Dir; COM to EVCC. IN_1 unused. Terminator cap off.

Phase A crossed per the Arctos drawing (not confirmed)



Source: Hardware/bench-wiring/generate_schematic.py
System wiring at connector level: Provisional items noted on the sheet
Karetao
Sheet: /
File: karetao_bench_wiring.kicad_sch

Title: Karetao bench wiring

Size: A2	Date: 2026-10-15	**Rev: 0.3**
KiCad E.D.A. 10.0.7		Id: 1/1